# DBMS Mini Project

Name – Dhanush S
SRN - PES1UG24AM360

Name – Darshan Govekar
SRN - PES1UG24AM355

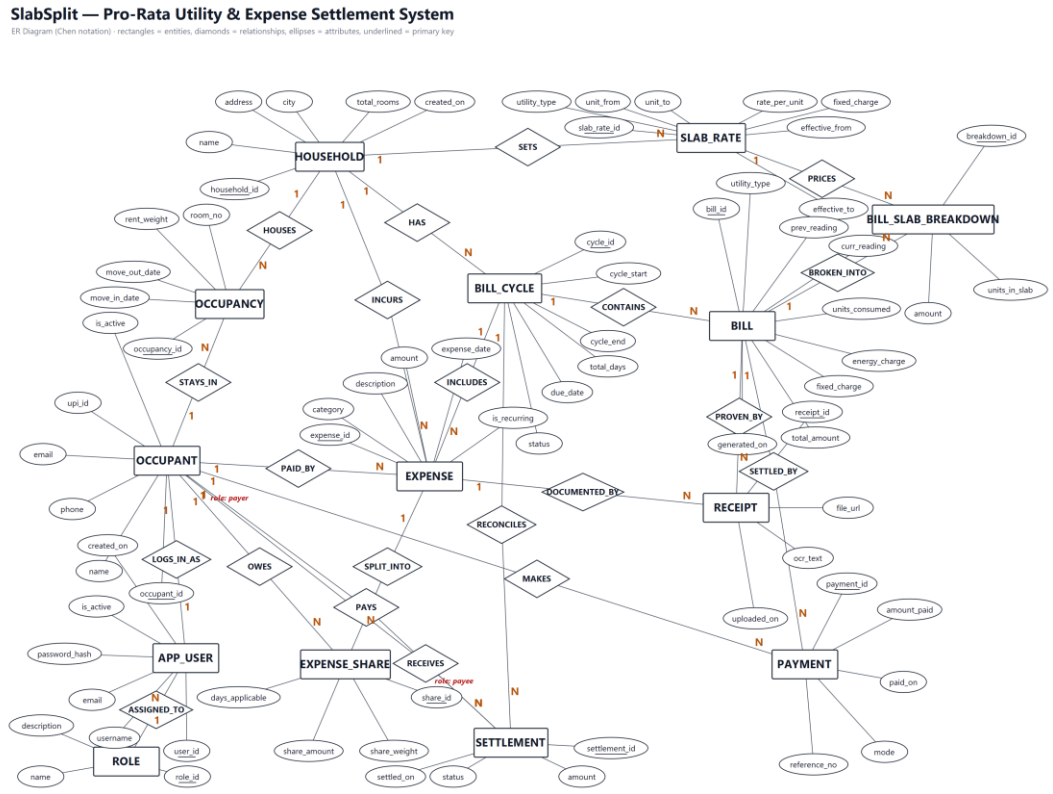
## Project title –

SlabSplit — Pro-Rata Utility & Expense Settlement System for Shared Households

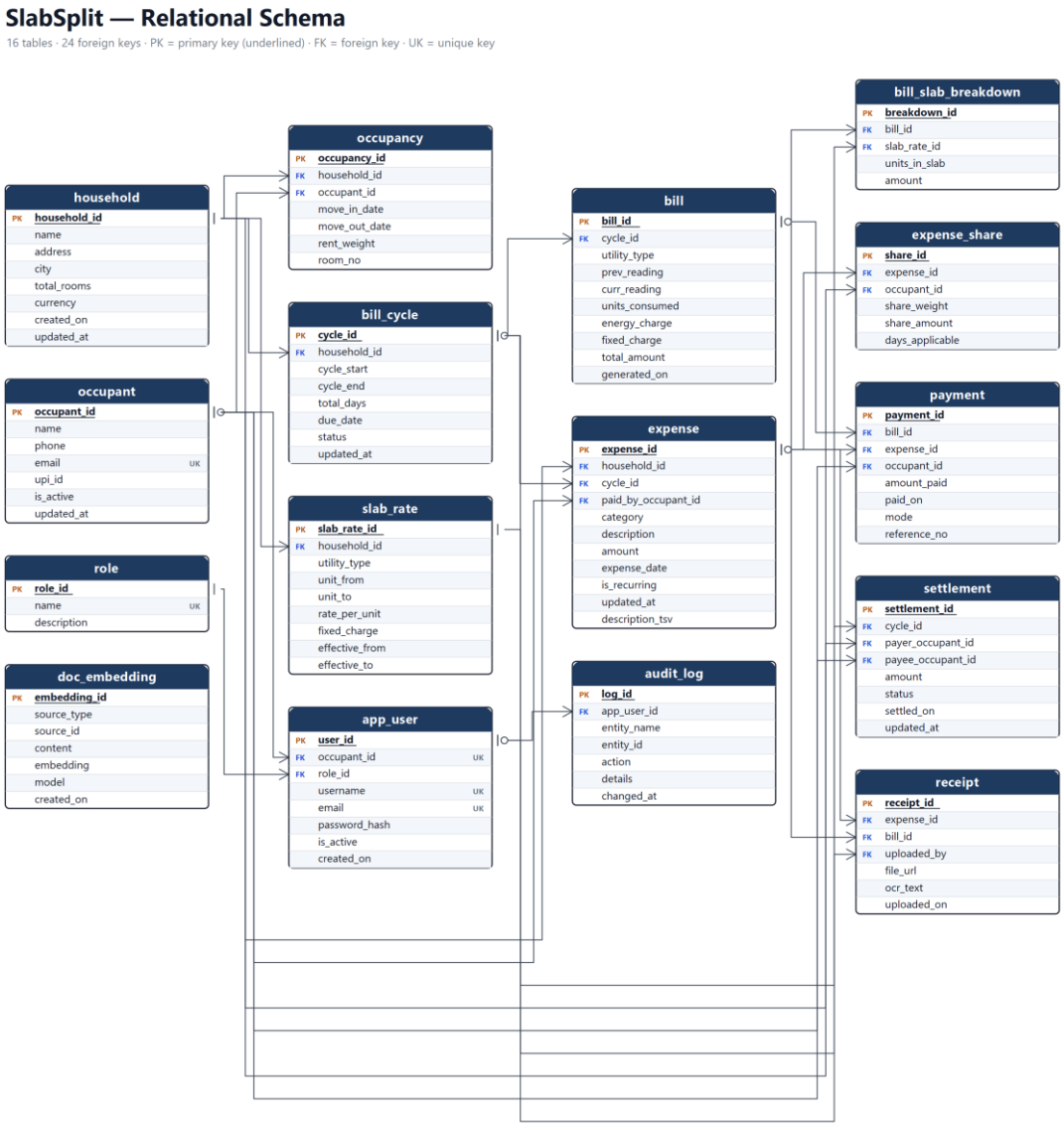
## Description

In shared flats, electricity is billed for the whole unit under a slab tariff and rent is split among current occupants, so an equal division becomes unfair when someone joins or vacates mid-cycle. SlabSplit stores occupancy periods, billing cycles, slab rates, and shared expenses in a centralized database, computes each occupant's day-weighted share, and nets pending dues into the minimum number of settlements between flatmates.

## ER Diagram



# Relational Schema




# DDL commands

CREATE EXTENSION IF NOT EXISTS vector;

CREATE TYPE utility_type_t AS ENUM ('electricity','water','gas','internet','maintenance');

CREATE TYPE cycle_status_t AS ENUM ('open','billed','closed');

CREATE TYPE payment_mode_t AS ENUM ('upi','cash','bank_transfer','card');

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CREATE TYPE settlement_status_t AS ENUM ('pending','settled','cancelled');

CREATE TYPE expense_category_t  AS ENUM
('utility','rent','internet','maintenance','groceries','repair','other');

CREATE TYPE doc_source_t       AS ENUM ('expense','receipt','settlement_note');


CREATE TABLE household (

   household_id  BIGSERIAL PRIMARY KEY,

   name        VARCHAR(120) NOT NULL,

   address      VARCHAR(255) NOT NULL,

   city        VARCHAR(80)  NOT NULL,

   total_rooms   SMALLINT    NOT NULL DEFAULT 1 CHECK (total_rooms BETWEEN 1
AND 50),

   currency     CHAR(3)     NOT NULL DEFAULT 'INR',

   created_on    DATE        NOT NULL DEFAULT CURRENT_DATE,

   updated_at    TIMESTAMPTZ  NOT NULL DEFAULT now()

);


CREATE TABLE occupant (

   occupant_id   BIGSERIAL PRIMARY KEY,

   name        VARCHAR(120) NOT NULL,

   phone       VARCHAR(20),

   email       VARCHAR(160) UNIQUE,

   upi_id      VARCHAR(80),

   is_active     BOOLEAN NOT NULL DEFAULT TRUE,

   updated_at    TIMESTAMPTZ NOT NULL DEFAULT now(),

   CONSTRAINT chk_occupant_email CHECK (email IS NULL OR email ~*
'^[^@\s]+@[^@\s]+\.[^@\s]+$'),

   CONSTRAINT chk_occupant_phone CHECK (phone IS NULL OR phone ~ '^[0-9+][0-9
\-]{6,19}$')
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);


CREATE TABLE occupancy (
   occupancy_id  BIGSERIAL PRIMARY KEY,
   household_id  BIGINT NOT NULL REFERENCES household(household_id) ON UPDATE CASCADE ON DELETE CASCADE,
   occupant_id   BIGINT NOT NULL REFERENCES occupant(occupant_id)  ON UPDATE CASCADE ON DELETE CASCADE,
   move_in_date  DATE   NOT NULL,
   move_out_date DATE,
   rent_weight   NUMERIC(5,2) NOT NULL DEFAULT 1.00 CHECK (rent_weight > 0),
   room_no      VARCHAR(20),
   CONSTRAINT chk_occupancy_dates CHECK (move_out_date IS NULL OR move_out_date >= move_in_date)
);


CREATE TABLE bill_cycle (
   cycle_id     BIGSERIAL PRIMARY KEY,
   household_id  BIGINT NOT NULL REFERENCES household(household_id) ON UPDATE CASCADE ON DELETE CASCADE,
   cycle_start   DATE NOT NULL,
   cycle_end     DATE NOT NULL,
   total_days    INT  GENERATED ALWAYS AS ((cycle_end - cycle_start) + 1) STORED,
   due_date     DATE,
   status       cycle_status_t NOT NULL DEFAULT 'open',
   updated_at    TIMESTAMPTZ NOT NULL DEFAULT now(),
   CONSTRAINT chk_cycle_range CHECK (cycle_end >= cycle_start),
   CONSTRAINT uq_cycle UNIQUE (household_id, cycle_start, cycle_end)
);
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CREATE TABLE slab_rate (
  slab_rate_id   BIGSERIAL PRIMARY KEY,
  household_id   BIGINT NOT NULL REFERENCES household(household_id) ON UPDATE CASCADE ON DELETE CASCADE,
  utility_type   utility_type_t NOT NULL,
  unit_from      NUMERIC(10,2) NOT NULL CHECK (unit_from >= 0),
  unit_to        NUMERIC(10,2),
  rate_per_unit  NUMERIC(10,4) NOT NULL CHECK (rate_per_unit >= 0),
  fixed_charge   NUMERIC(10,2) NOT NULL DEFAULT 0 CHECK (fixed_charge >= 0),
  effective_from DATE NOT NULL,
  effective_to   DATE,
  CONSTRAINT chk_slab_bounds    CHECK (unit_to IS NULL OR unit_to > unit_from),
  CONSTRAINT chk_slab_eff_dates CHECK (effective_to IS NULL OR effective_to >= effective_from)
);

CREATE TABLE bill (
  bill_id       BIGSERIAL PRIMARY KEY,
  cycle_id      BIGINT NOT NULL REFERENCES bill_cycle(cycle_id) ON UPDATE CASCADE ON DELETE CASCADE,
  utility_type   utility_type_t NOT NULL,
  prev_reading   NUMERIC(12,2) NOT NULL DEFAULT 0 CHECK (prev_reading >= 0),
  curr_reading   NUMERIC(12,2) NOT NULL CHECK (curr_reading >= 0),
  units_consumed NUMERIC(12,2) GENERATED ALWAYS AS (curr_reading - prev_reading) STORED,
  energy_charge  NUMERIC(12,2) NOT NULL DEFAULT 0 CHECK (energy_charge >= 0),
  fixed_charge   NUMERIC(12,2) NOT NULL DEFAULT 0 CHECK (fixed_charge >= 0),
  total_amount   NUMERIC(12,2) NOT NULL DEFAULT 0 CHECK (total_amount >= 0),
```

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  generated_on   DATE NOT NULL DEFAULT CURRENT_DATE,
  CONSTRAINT chk_bill_reading CHECK (curr_reading >= prev_reading),
  CONSTRAINT uq_bill UNIQUE (cycle_id, utility_type)
);

CREATE TABLE bill_slab_breakdown (
  breakdown_id  BIGSERIAL PRIMARY KEY,
  bill_id      BIGINT NOT NULL REFERENCES bill(bill_id)         ON UPDATE CASCADE ON DELETE CASCADE,
  slab_rate_id  BIGINT NOT NULL REFERENCES slab_rate(slab_rate_id) ON UPDATE CASCADE ON DELETE RESTRICT,
  units_in_slab NUMERIC(12,2) NOT NULL CHECK (units_in_slab >= 0),
  amount       NUMERIC(12,2) NOT NULL CHECK (amount >= 0),
  CONSTRAINT uq_bill_slab UNIQUE (bill_id, slab_rate_id)
);

CREATE TABLE expense (
  expense_id        BIGSERIAL PRIMARY KEY,
  household_id      BIGINT NOT NULL REFERENCES household(household_id) ON UPDATE CASCADE ON DELETE CASCADE,
  cycle_id          BIGINT REFERENCES bill_cycle(cycle_id) ON UPDATE CASCADE ON DELETE SET NULL,
  paid_by_occupant_id BIGINT NOT NULL REFERENCES occupant(occupant_id) ON UPDATE CASCADE ON DELETE RESTRICT,
  category          expense_category_t NOT NULL DEFAULT 'other',
  description       TEXT NOT NULL,
  amount            NUMERIC(12,2) NOT NULL CHECK (amount > 0),
  expense_date      DATE NOT NULL DEFAULT CURRENT_DATE,
  is_recurring      BOOLEAN NOT NULL DEFAULT FALSE,
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  updated_at        TIMESTAMPTZ NOT NULL DEFAULT now(),
  description_tsv   tsvector GENERATED ALWAYS AS (to_tsvector('english', coalesce(description,''))) STORED
);

CREATE TABLE expense_share (
  share_id       BIGSERIAL PRIMARY KEY,
  expense_id     BIGINT NOT NULL REFERENCES expense(expense_id)  ON UPDATE CASCADE ON DELETE CASCADE,
  occupant_id    BIGINT NOT NULL REFERENCES occupant(occupant_id) ON UPDATE CASCADE ON DELETE CASCADE,
  share_weight   NUMERIC(6,3) NOT NULL DEFAULT 1.000 CHECK (share_weight >= 0),
  share_amount   NUMERIC(12,2) NOT NULL CHECK (share_amount >= 0),
  days_applicable INT NOT NULL DEFAULT 0 CHECK (days_applicable >= 0),
  CONSTRAINT uq_expense_share UNIQUE (expense_id, occupant_id)
);

CREATE TABLE payment (
  payment_id   BIGSERIAL PRIMARY KEY,
  bill_id      BIGINT REFERENCES bill(bill_id)      ON UPDATE CASCADE ON DELETE SET NULL,
  expense_id   BIGINT REFERENCES expense(expense_id) ON UPDATE CASCADE ON DELETE SET NULL,
  occupant_id  BIGINT NOT NULL REFERENCES occupant(occupant_id) ON UPDATE CASCADE ON DELETE RESTRICT,
  amount_paid  NUMERIC(12,2) NOT NULL CHECK (amount_paid > 0),
  paid_on      DATE NOT NULL DEFAULT CURRENT_DATE,
  mode         payment_mode_t NOT NULL DEFAULT 'upi',
  reference_no VARCHAR(80),
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  CONSTRAINT chk_payment_target CHECK (bill_id IS NOT NULL OR expense_id IS NOT NULL)
);

CREATE TABLE settlement (
  settlement_id     BIGSERIAL PRIMARY KEY,
  cycle_id          BIGINT NOT NULL REFERENCES bill_cycle(cycle_id) ON UPDATE CASCADE ON DELETE CASCADE,
  payer_occupant_id BIGINT NOT NULL REFERENCES occupant(occupant_id) ON UPDATE CASCADE ON DELETE RESTRICT,
  payee_occupant_id BIGINT NOT NULL REFERENCES occupant(occupant_id) ON UPDATE CASCADE ON DELETE RESTRICT,
  amount            NUMERIC(12,2) NOT NULL CHECK (amount > 0),
  status            settlement_status_t NOT NULL DEFAULT 'pending',
  settled_on        DATE,
  updated_at        TIMESTAMPTZ NOT NULL DEFAULT now(),
  CONSTRAINT chk_settle_distinct CHECK (payer_occupant_id <> payee_occupant_id)
);

CREATE TABLE role (
  role_id     SMALLSERIAL PRIMARY KEY,
  name        VARCHAR(30) NOT NULL UNIQUE,
  description VARCHAR(160)
);

CREATE TABLE app_user (
  user_id     BIGSERIAL PRIMARY KEY,
  occupant_id BIGINT UNIQUE REFERENCES occupant(occupant_id) ON UPDATE CASCADE ON DELETE SET NULL,
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  role_id      SMALLINT NOT NULL REFERENCES role(role_id) ON UPDATE CASCADE ON DELETE RESTRICT,
  username     VARCHAR(60)  NOT NULL UNIQUE,
  email        VARCHAR(160) NOT NULL UNIQUE,
  password_hash VARCHAR(255) NOT NULL,
  is_active    BOOLEAN NOT NULL DEFAULT TRUE,
  created_on   TIMESTAMPTZ NOT NULL DEFAULT now(),
  CONSTRAINT chk_user_hash_not_plain CHECK (char_length(password_hash) >= 20)
);

CREATE TABLE receipt (
  receipt_id  BIGSERIAL PRIMARY KEY,
  expense_id  BIGINT REFERENCES expense(expense_id) ON UPDATE CASCADE ON DELETE CASCADE,
  bill_id     BIGINT REFERENCES bill(bill_id)      ON UPDATE CASCADE ON DELETE CASCADE,
  uploaded_by BIGINT REFERENCES app_user(user_id)   ON UPDATE CASCADE ON DELETE SET NULL,
  file_url    VARCHAR(500) NOT NULL,
  ocr_text    TEXT,
  uploaded_on TIMESTAMPTZ NOT NULL DEFAULT now(),
  CONSTRAINT chk_receipt_target CHECK (expense_id IS NOT NULL OR bill_id IS NOT NULL)
);

CREATE TABLE doc_embedding (
  embedding_id BIGSERIAL PRIMARY KEY,
  source_type  doc_source_t NOT NULL,
  source_id    BIGINT NOT NULL,
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  content     TEXT   NOT NULL,
  embedding   vector(384) NOT NULL,
  model       VARCHAR(60) NOT NULL DEFAULT 'all-MiniLM-L6-v2',
  created_on  TIMESTAMPTZ NOT NULL DEFAULT now(),
  CONSTRAINT uq_doc_source UNIQUE (source_type, source_id)
);

CREATE TABLE audit_log (
  log_id     BIGSERIAL PRIMARY KEY,
  app_user_id BIGINT REFERENCES app_user(user_id) ON UPDATE CASCADE ON
DELETE SET NULL,
  entity_name VARCHAR(40) NOT NULL,
  entity_id   BIGINT,
  action     VARCHAR(10) NOT NULL CHECK (action IN ('INSERT','UPDATE','DELETE')),
  details    JSONB,
  changed_at  TIMESTAMPTZ NOT NULL DEFAULT now()
);
```